PRAKTIKUM SISTEM PENGENDALI ELEKTRIK

FLOWCHART TERPADU FIRMWARE HOVERBOARD-FIRMWARE-HACK-FOC | REVERSE ENGINEERING

Satu kanvas: build/flash, input variants, mixer, tiga control type, dual motor PWM, fault protection, serta feedback UART.

hoverboard-robotics/hoverboard-firmware-hack-FOC | branch master | STM32F103 / GD32 | Dual BLDC FOC

00 | ARSITEKTUR FIRMWARE HOVERBOARD FOC

Sideboard kiri/kanan
USART2 / USART3

Input pengguna / receiver
ADC, UART, PWM, PPM, iBUS, Nunchuk

command / feedback

sinyal input

Tegangan baterai & suhu board
ADC, filter, kalibrasi

MCU utama
STM32F103RCT6 / GD32F103RCT6

Vbat / suhu / DC current

raw input + measurements

Aplikasi utama
main.c + util.c
input mapping & supervision

Feedback arus dan bus DC
ADC1 / ADC2 + adc_buffer

debug / buzzer / telemetry

pwml / pwmr + enable

phase current

UART feedback / debug
LED, buzzer, LCD opsional

BLDC_Controller
controller model per motor kiri/kanan

duty / phase command

Timer PWM 3-fasa
LEFT / RIGHT

switching pulses

Power bridge MOSFET
phase U / V / W

feedback Hall

tegangan 3-fasa

Motor BLDC hoverboard
LEFT + RIGHT

posisi / speed

Sensor Hall motor
posisi rotor & kecepatan

04 | DUAL BLDC CONTROL: COMMUTATION / SIN / FOC

Terima pwml / pwmr
input reference motor LEFT / RIGHT

enable = 1 dan tidak ada z_errCode?

YA

TIDAK

CTRL_TYP_SEL?

ADC current feedback

SIN_CTRL

FOC_CTRL

COM_CTRL

SIN_CTRL
tentukan electrical angle dari Hall

FOC: baca phase-current feedback
dan posisi / speed Hall

Estimasi sudut listrik, arah
dan kecepatan motor

Transformasi Clarke / Park
Iabc -> Ialpha/beta -> Id / Iq

Nonaktifkan drive / target nol
PWM output tidak mendorong motor

CTRL_MOD_REQ?

COM_CTRL
komutasi berbasis state Hall
phase sequence

OPEN_MODE

VLT_MODE

SPD_MODE

TRQ_MODE

OPEN_MODE
perintah terbuka tanpa loop speed

VLT_MODE
referensi tegangan motor

SPD_MODE
loop speed menghasilkan referensi arus

TRQ_MODE
referensi torque / current

Sinusoidal modulation
phase advance bila diaktifkan

open reference

voltage reference

speed loop

torque reference

Regulator arus d/q
PI dan limit referensi

Inverse Park / tegangan alpha-beta
SVPWM menghasilkan duty U/V/W

phase output

output off

FIELD_WEAK_ENA aktif?

YA

TIDAK

FOC: field weakening
SIN: phase advance

field / phase adjustment

Terapkan limit kecepatan dan arus
N_MOT_MAX / I_MOT_MAX
COM/SIN: chopping sesuai I_DC_MAX

Update timer compare / PWM 3-fasa
LEFT & RIGHT motor

Siklus kontrol berikutnya

battery and board temperature

05 | SAFETY SUPERVISION, LIMIT, FAULT & POWEROFF

Batas runtime sesuai config.h
I_MOT_MAX, I_DC_MAX, N_MOT_MAX
TIMEOUT input

Siklus pemeriksaan keselamatan

parameter

Periksa z_errCode kiri/kanan
ADC timeout, serial timeout, general timeout

Ada error controller motor?

YA

TIDAK

Input hilang / invalid / timeout?

YA

TIDAK

Set enable = 0
nonaktifkan penggerak dan beri beep error

Periksa BAT_DEAD dan TEMP_POWEROFF
serta speedAvgAbs

Set timeout flag dan peringatan
zero command / disable sesuai variant

Baterai kritis atau overtemperature
dan speedAvgAbs < 20?

TIDAK

YA

Tombol power meminta shutdown?

TIDAK

YA

TRANSPOTTER: error jarak > batas aman?

TIDAK

YA

Peringatan nonfatal
battery level, temperature, reverse drive

poweroff()
matikan power latch / shutdown

shutdown

Kembali ke main loop
atau pertahankan drive disabled

motor enable flag

controller disabled

Hall position / speed feedback

input timeout flag

UART feedback output

PWM to power bridge

fault and warning signaling

03 | MAIN LOOP, FILTER, MIXER & COMMAND MOTOR

while(1)
main.c

Waktu loop cukup?
DELAY_IN_MAIN_LOOP

YA

TIDAK

Panggil readCommand()

Lanjutkan loop / tunggu tick berikutnya

calcAvgSpeed()
hitung speedAvg dan speedAvgAbs

Motor boleh di-enable?
input awal mendekati nol
dan kedua motor bebas error

TIDAK

YA

Pertahankan motor disabled
sampai syarat start terpenuhi

enable = 1
reset filter dan bunyikan konfirmasi

command tetap dibatasi

rateLimiter16()
batasi perubahan throttle / steering

filtLowPass32()
filter speed dan steer

Logika variant aktif?
HOVERCAR / SKATEBOARD / TRANSPOTTER

TRANSPOTTER

HOVERCAR / SKATEBOARD

TRANSPOTTER: ukur Gametrak distance
steering + target distance

default / lainnya

Brake blend dan reverse gate
MultipleTapBrake / speedBlend

cmdL / cmdR

speed / steer

mixerFcn(speed, steer, &cmdR, &cmdL)
Differential drive mixing

Terapkan INVERT_R_DIRECTION
dan INVERT_L_DIRECTION

Set pwmr dan pwml
range perintah kira-kira -1000..1000

Update power-button check, buzzer,
battery, temperature dan diagnostics

Kembali ke awal loop

06 | SENSOR, DIAGNOSTIK & SERIAL FEEDBACK

Mulai update pengukuran / status

Ambil ADC buffer
batt1, temp, arus DC, speed motor, odometri

Filter temperatur board
filtLowPass32() + konversi ADC ke °C

Kalibrasi tegangan baterai
BAT_CALIB_REAL_VOLTAGE / BAT_CALIB_ADC

Hitung arus DC kiri, kanan
dan total DC link current

Proses data sideboard
LED dan sensor (jika aktif)

DEBUG_SERIAL aktif dan interval tercapai?
setiap ~125 ms

YA

TIDAK

Cetak command, raw ADC,
Vbat dan temperatur ke UART

FEEDBACK_SERIAL aktif dan interval?
setiap ~10 ms

YA

TIDAK

Susun frame feedback
command, speed L/R, odom, Vbat, suhu, LED

Hitung checksum XOR frame

Kirim frame via USART2/3 TX DMA

Atur pola buzzer / LED
error, timeout, suhu, baterai, reverse

LCD enabled

LCD opsional: status / jarak / baterai

Lanjut loop berikutnya

02 | AKUISISI INPUT, DECODE & INPUT VALIDATION

readCommand()
peroleh input1[] dan input2[]

Sumber input aktif?

ADC

USART

PWM

PPM

iBUS

Nunchuk

Sideboard

ADC analog 12-bit
potensiometer / pedal

USART2 / USART3
command serial via DMA

PWM receiver
ukur pulse width timer

PPM-Sum receiver
decode channel capture

FlySky iBUS
decode channel data

Wii Nunchuk via I2C
joystick / buttons

Sideboard serial data
sensor / command fields

Decode / baca nilai mentah
input1[inIdx].raw, input2[inIdx].raw

Dual-input aktif?
Pilih primary / auxiliary sesuai prioritas

primary / auxiliary

Terapkan MIN / MID / MAX, deadband,
normalisasi dan opsi invert

Data masuk valid dan baru?

YA

TIDAK

Set timeout flag sesuai sumber
ADC / Serial / General (PPM, PWM, Nunchuk)

Perbarui input1[inIdx].cmd
dan input2[inIdx].cmd

restart only after power-up / reset

Gunakan input primary bila aux terputus
atau nilai aman sesuai variant

Kembalikan command terukur
ke main loop

readCommand() call

pwml / pwmr target

input commands returned

runtime safety monitoring

battery / temperature / status

input subsystem initialized

startup complete

selected FOC control mode

current / speed limits

01 | BUILD, FLASH & INISIALISASI MCU

MULAI: siapkan source firmware

Pilih satu VARIANT_*
di config.h / platformio.ini

Atur CTRL_TYP_SEL, CTRL_MOD_REQ,
input pin dan limit keselamatan

Konfigurasi target valid?

ulang konfigurasi

YA

TIDAK

Compile via PlatformIO / Keil / Make

Build gagal / konflik define
perbaiki konfigurasi lalu build ulang

build sukses

Hasil build firmware (.hex / .bin)

Flash via SWD / ST-Link
GND, 3V3, SWDIO, SWCLK
Jangan supply board dari 3V3 programmer

perhatikan keselamatan

Saat flashing: tahan tombol power
dan siapkan baterai >36 V

RESET / entry main()

HAL_Init() dan set prioritas NVIC

SystemClock_Config()
aktifkan clock sistem

MX_GPIO_Init()
MX_TIM_Init() + MX_ADC1/2_Init()

BLDC_Init()
inisialisasi model controller motor

Aktifkan power latch
OFF_PORT / OFF_PIN = SET

Input_Lim_Init() + Input_Init()

Mulai ADC1 dan ADC2
isi awal pengukuran board temperature

Bunyikan power-on melody
LED status ON

Tombol power masih ditekan?

YA

periksa lagi

TIDAK

Tunggu tombol dilepas
HAL_Delay(10 ms)

Masuk while(1)
main loop berjalan

07 | VARIANT & PARAMETER DEFAULT (config.h)

I_MOT_MAX = 15 A; I_DC_MAX = 17 A
N_MOT_MAX = 1000 rpm
MOTOR_LEFT_ENA + MOTOR_RIGHT_ENA

PWM_FREQ = 16000 Hz; DEAD_TIME = 48
DELAY_IN_MAIN_LOOP = 5 ms (TRANSPOTTER: 2 ms)
TIMEOUT = 20

CTRL_MOD_REQ: OPEN / VOLTAGE / SPEED / TORQUE
Default: VLT_MODE

CTRL_TYP_SEL: COM_CTRL / SIN_CTRL / FOC_CTRL
Default: FOC_CTRL

VARIANT_ADC / USART / NUNCHUK / PPM / PWM / iBUS
HOVERCAR / HOVERBOARD / TRANSPOTTER / SKATEBOARD

compile-time selected input path

timing / input timeout settings

selected motor-control algorithm

Sumber: README.md, main.c, util.c, setup.c, BLDC_controller.c, dan config.h dari hoverboard-robotics/hoverboard-firmware-hack-FOC (master). FOC diringkas sebagai alur kontrol.

SATU KANVAS | 124 NODE | 165 KONEKSI